Quantum Algorithms with Qiskit

An Open-Source Framework for Quantum Computing

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What is Qiskit?

Build, simulate, and run quantum circuits on real IBM quantum hardware via the cloud.

Open-source Python SDK, commonly used alongside NumPy and Matplotlib for scientific computing workflows.

Hardware-agnostic circuit construction with an extensive standard library of gates and algorithms.

Python SDK

NumPy

Matplotlib

Qiskit-Aer

Core Architecture

Qiskit Core

Circuit construction
Transpilation
Quantum info

Qiskit-Aer

High-performance
local simulators
Noise modeling

Qiskit-Algorithms

Grover
Eigensolver
NumPyEigensolver

Circuits and Registers

`QuantumCircuit` — the primary class used to assemble gates. Initialize a number of qubits and classical bits, then sequentially add gates.

Registers

Allow better organization of qubits or classical bits into named groups for complex circuit designs.

Example

```
qc = QuantumCircuit(2, 1)
```

2 qubits, 1 classical bit

Quantum Gates

Pauli Gates

```
qc.x()  qc.y()  qc.z()
```

Bit-flip, phase, and combined operations

Superposition

```
qc.h()
```

Hadamard — creates equal superposition

Entanglement

```
qc.cx(control, target)
```

Controlled-NOT for two-qubit entanglement

Composability: Build sub-circuits as reusable gates with `.to_gate()` and `.control()`

Other Useful Features

Barriers

Visually separate parts of the circuit diagram and act as a compiler directive to prevent optimization across the boundary.

```
qc.barrier()
```

Measurement

Collapses a quantum state into a classical 0 or 1, storing the result in the specified classical bit.

```
qc.measure(qubit, bit)
```

Execution

1

Transpile

Translate abstract circuits to native hardware gate sets the simulator or QPU supports.

2

Simulate

AerSimulator emulates quantum execution on the CPU — without noise or errors of current QPUs.

3

Sample

Specify the number of shots to run the circuit and collect a probability distribution of results.

Visualizing Results

Circuit Diagrams

Qiskit renders circuits using Matplotlib for clear, publication-quality visualizations of gate sequences.

```
qc.draw('mpl')
```

Outcome Histograms

Visualize the probability distribution of measurement outcomes from executing a circuit across many shots.

```
plot_histogram(counts)
```

Demo